SCAPS-1D / LEAD-FREE PEROVSKITE PHOTOVOLTAICS

Numerical Optimization of Lead-Free $FASnI_3$ Perovskite Solar Cells via SCAPS-1D Simulation

Thickness, defect and interface engineering in an n–i–p perovskite solar cell

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Manuscript under review

Device Architecture & Energy-Band Alignment

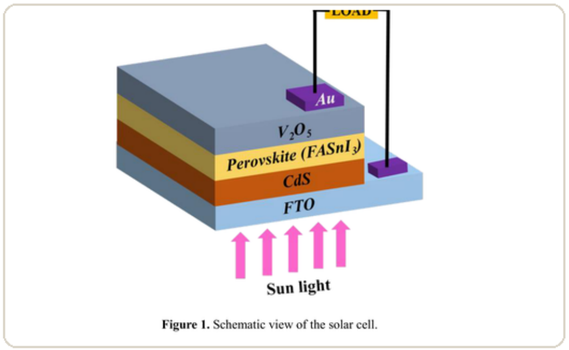


Figure 1. Schematic view of the solar cell.

Fig. 1 · $FASnI_3$ manuscript · PDF p.7

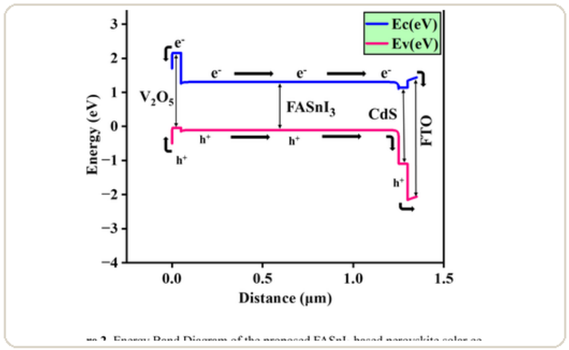
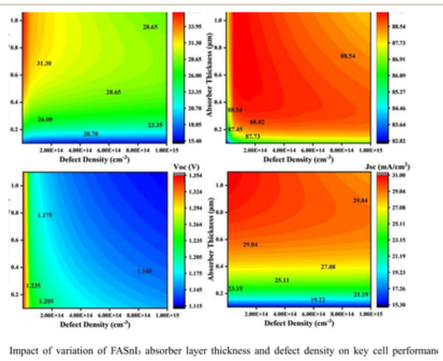


Fig. 2 · $FASnI_3$ manuscript · PDF p.8

n–i–p stack: FTO/CdS/$FASnI_3$/V_2O_5/Au. CdS = ETL; V_2O_5 = HTL.

Original manuscript band diagram is reproduced; no additional band-edge values are inferred.

Absorber Thickness & Bulk-Defect Optimization



Impact of variation of $FASnI_3$ absorber layer thickness and defect density on key cell performan

Fig. 3 · $FASnI_3$ manuscript · PDF p.9

Original SCAPS-1D contour maps show PCE, FF, Voc and Jsc over absorber thickness 0.1–1.0 μm and bulk defect density 2×10^{14}–1×10^{15} cm^{-3}.

Design window: absorber 0.8–1.0 μm with bulk defects below 4×10^{14} cm^{-3}.

Optimized Simulated Device Performance

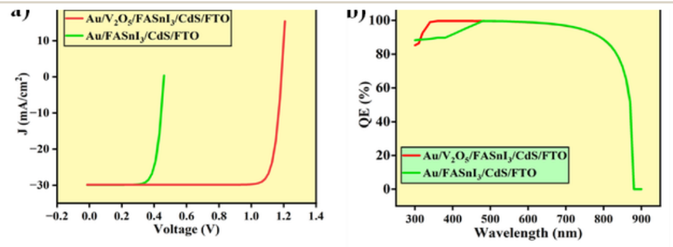


Figure 10. (a) J-V, and (b) QE analysis curve of the proposed solar cell.

The QE curve (**Figure 10(b)**) shows the spectral response, indicating the percentage of incident photons onverted into collected electrons at different wavelengths. The red curve (FTO/CdS/$FASnI_3$/V_2O_5/Au) exhibits a onsistently high quantum efficiency (approaching 100%) across a broad range of the spectrum, particularly from

Fig. 10 · $FASnI_3$ manuscript · PDF p.18

PCE 30.92%; Voc 1.19 V; Jsc 29.88 mA cm^{-2}; FF 87.18%.

Optimized thicknesses: $FASnI_3$ 800 nm; CdS 50 nm; V_2O_5 50 nm. Baseline conditions: AM1.5, 300 K.

J–V and QE curves reproduced from the manuscript; all performance metrics are simulated.

Transport-Layer & Interface Design Windows

CdS ETL: thickness below 0.2 μm; bulk defect density below 6×10^{16} cm^{-3}.

V_2O_5 HTL: thickness below 0.2 μm; bulk defect density below 4×10^{16} cm^{-3}.

Both absorber/transport-layer interfaces: defect density below 4×10^{13} cm^{-2}.

These are simulated design windows; experimental fabrication and validation remain future work.

MANUSCRIPT STATUS Under review. Device-performance values shown are SCAPS-1D numerical predictions.